

# Requirement Analysis

## Technology Stack (Architecture & Stack)

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 08 September 2026                                  |
| Student Name  | Avanapu Roopa                                      |
| Project Name  | Automated Network Request Management in ServiceNow |
| Maximum Marks |                                                    |

### Technical Architecture

#### Technology Stack & Layered Architecture

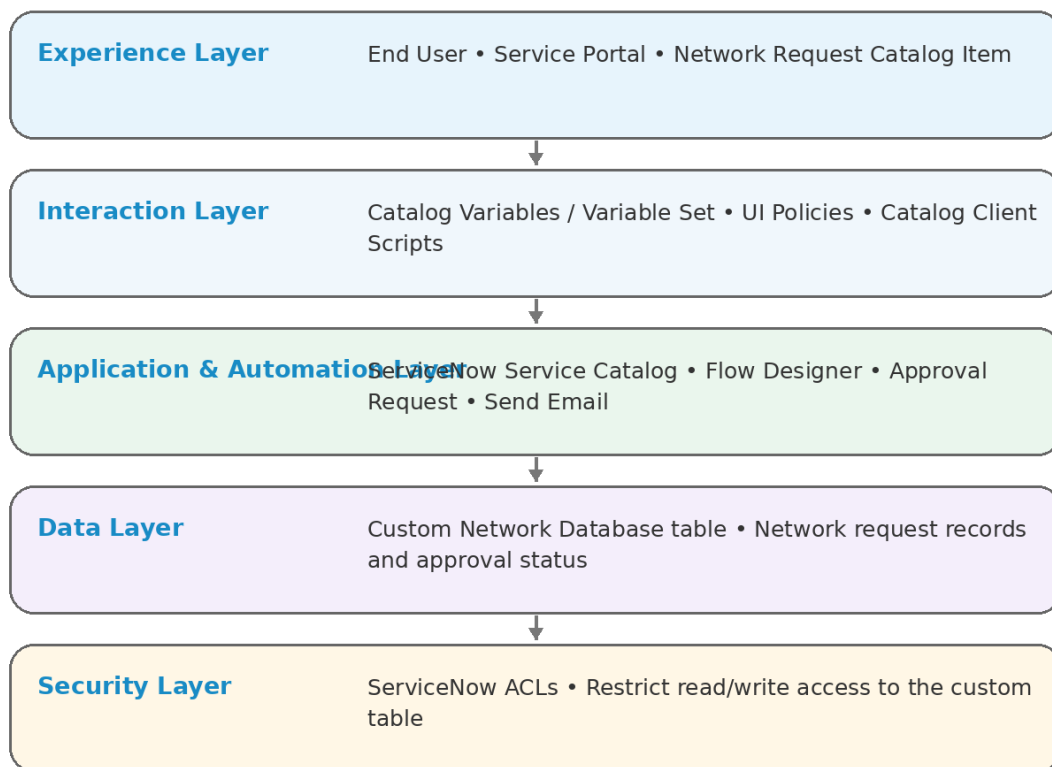


Figure 1: Layered technology architecture for the Automated Network Request Management solution.

The solution is implemented within ServiceNow using the Service Portal, Service Catalog, catalog variables and variable sets, Flow Designer, a custom Network Database table, approval requests, UI Policies, catalog client scripts, and ACLs. No external API, machine-learning model, or separate cloud database is specified in the supplied project instructions.

Figure 1: ServiceNow technology stack and request-management architecture flow.

**Table-1: Components & Technologies**

| S.No | Component                       | Description                                                                                                                                                                                                 | Technology                                              |
|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1    | User Interface                  | Service Portal through which end users search for Network Requests, fill required details, and submit requests.                                                                                             | ServiceNow Service Portal                               |
| 2    | Application Logic-1             | Catalog request handling and variable configuration.                                                                                                                                                        | ServiceNow Service Catalog                              |
| 3    | Application Logic-2             | End-to-end request lifecycle automation including Get Catalog Variables, Create Record, Send Email, Ask for Approval, If/Else, and Update Record.                                                           | ServiceNow Flow Designer                                |
| 4    | Application Logic-3             | Dynamic field visibility and dependent form behaviour.                                                                                                                                                      | Catalog UI Policies and Catalog Client Scripts          |
| 5    | Database                        | Stores network request records and fields including Database Number, Assignment To, Mobile Number, Total Amount, Mode of Payment, Type of Connection, Customer Address, Approval Status, and Requested For. | ServiceNow custom Network Database table                |
| 6    | Approval                        | Stores/displays approval request information related to the Network Database record.                                                                                                                        | ServiceNow Approval Request / sysapprovals relationship |
| 7    | Email Notification              | Communicates request progress from the automated flow.                                                                                                                                                      | ServiceNow Flow Designer Send Email action              |
| 8    | External API-1                  | No external API is specified in the project instructions.                                                                                                                                                   | Not specified                                           |
| 9    | Machine Learning Model          | No machine-learning model is specified in the project instructions.                                                                                                                                         | Not specified                                           |
| 10   | Infrastructure (Server / Cloud) | Development and testing are performed in a ServiceNow Personal Developer Instance (PDI).                                                                                                                    | ServiceNow PDI                                          |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                                                                    | Technology                                |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1    | Open-Source Frameworks   | No external open-source framework is specified.                                                                                                | Not specified                             |
| 2    | Security Implementations | Default ACLs on the custom table are used to restrict read/write access according to configured access controls.                               | ServiceNow ACLs                           |
| 3    | Scalable Architecture    | The solution uses a structured custom table and automated Flow Designer process; no specific microservices or scaling technology is specified. | ServiceNow platform                       |
| 4    | Availability             | The solution runs within the ServiceNow platform/PDI; no separate availability infrastructure is specified.                                    | ServiceNow platform                       |
| 5    | Performance              | Project objectives include reduced manual effort and faster request fulfilment; no requests-per-second, caching, or CDN metric is specified.   | ServiceNow Flow Designer / Service Portal |